

|    |   |                                                                                                                                                                |
|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | Option D: zero velocity and positive acceleration                                                                                                              |
| 14 | A | $E = \frac{1}{2}m\omega^2 x^2$ <p>When <math>x = \frac{x_0}{4}</math>, <math>PE = \frac{1}{2}m\omega^2 \left(\frac{x_0}{4}\right)^2 = \frac{1}{16}E</math></p> |
| 15 | B | I                                                                                                                                                              |